SAEON Rainfall Monitoring User Stories and Tasks

Project Overview

The SAEON Real-Time Rainfall Monitoring System uses an ESP32 WROOM-32, TR-525M rain gauge, SHT31 temperature and humidity sensor and SIM7000G cellular module. Readings pass through HiveMQ Cloud to Firebase and the dashboard. The 13 week plan runs from 28 July to 23 October 2026, Tuesday to Friday. Historical data, CSV export, authenticated access, station diagnostics and solar/battery integration are included in scope. US12 deployment and field-validation activities remain In Progress; GPS, extra weather sensors and over-the-air updates are future enhancements. Trello tracks work and GitHub stores code and evidence. These stories define acceptance requirements, not additional claims of field completion.

US01 Monitor Rainfall

User story

As a SAEON researcher, I want the station to measure rainfall automatically so that I can collect observations without manually retrieving a USB data logger.

Acceptance criteria

- The TR-525M rain gauge produces a recorded count for each bucket tip.
- Contact bounce does not produce duplicate counts during controlled testing.
- The system converts counts into rainfall depth in millimetres using the verified calibration factor for the installed gauge.

Tasks

- Connect the rain gauge to the ESP32.
- Implement tip counting and debouncing.
- Implement rainfall calculations.
- Compare recorded results with controlled test measurements.

US02 Monitor Temperature

User story

As a SAEON researcher, I want temperature readings alongside rainfall data so that I can understand the environmental conditions at the station.

Acceptance criteria

- The SHT31 provides temperature readings in degrees Celsius.
- Readings are checked against a reference thermometer.
- Missing or invalid sensor readings are identified rather than displayed as valid measurements.

Tasks

- Connect the temperature sensor.
- Implement sensor reading and error handling.
- Compare measurements with a reference thermometer.
- Include temperature in the station's data message.

US03 Transmit Data Remotely

User story

As a SAEON researcher, I want the station to transmit readings through a cellular network so that I can monitor remote sites without relying on local Wi-Fi.

Acceptance criteria

- The SIM7000G establishes a data connection using the configured SIM and network settings.
- Readings reach HiveMQ Cloud through MQTT.
- Each message includes a station identifier, timestamp, and clearly defined measurement fields.
- Connection failures are recorded, and the station attempts to reconnect.

Tasks

- Configure SIM7000G wiring and power.
- Configure cellular network access.
- Configure MQTT topics and authentication.
- Test publishing and reconnection.

US04 Store Monitoring Records

User story

As a SAEON researcher, I want readings saved in a database so that I can review environmental observations over time.

Acceptance criteria

- Valid incoming readings are stored in Firebase.
- Each reading is associated with the correct station and observation time.
- Invalid messages are rejected or recorded as errors.
- Stored records can be retrieved by the dashboard.

Tasks

- Create Firebase collections for stations and telemetry.
- Develop the MQTT-to-database bridge.
- Validate incoming messages.
- Test storage and retrieval.

US05 View Current Conditions

User story

As a SAEON researcher, I want a dashboard showing the latest rainfall and temperature readings so that I can check current conditions remotely.

Acceptance criteria

- The dashboard displays rainfall and temperature with their units.
- The station identifier and latest update time are visible.
- New readings update the displayed information.
- Missing or outdated data is clearly identified.

Tasks

- Create the dashboard layout.
- Connect the dashboard to monitoring data.
- Implement live updates.
- Add loading, error, and stale-data displays.

US06 Review Historical Data

User story

As a SAEON researcher, I want to filter historical readings by station and date so that I can examine changes and compare observations.

Acceptance criteria

- Users can select a station and date range.
- Charts and tables show records matching the selected filters.
- Rainfall and temperature are labelled with the correct units.
- An empty result is clearly explained.

Tasks

- Implement historical data queries.
- Add station and date filters.
- Create rainfall and temperature charts.
- Test filters against stored records.

US07 Export Monitoring Data

User story

As a SAEON researcher, I want to export selected readings as a CSV file so that I can analyse them using other tools.

Acceptance criteria

- The export matches the selected station and date range.
- Column headings identify measurements and units.
- Exported values match the records displayed in the dashboard.
- Timestamps use a documented time zone.

Tasks

- Create the CSV export function.
- Apply the dashboard filters to exported data.
- Add clear column headings.
- Compare an export with database records.

US08 Check Station Health

User story

As a station operator, I want to view station health information so that I can identify power or communication problems.

Acceptance criteria

- The dashboard shows the latest available battery reading and last communication time.
- Missing health information is clearly identified.
- Low battery and stale communication are flagged using agreed thresholds.
- Old readings are not presented as evidence that the station is currently healthy.

Tasks

- Implement battery measurement and calibration.
- Include diagnostic fields in transmitted messages.
- Create the station health panel.
- Test low-power and connection-loss displays.

US09 Access the Dashboard Securely

User story

As an authorised SAEON user, I want to sign in securely so that monitoring information and restricted functions are available according to my access permissions.

Acceptance criteria

- Valid users can sign in and sign out.
- Restricted pages require an authenticated session.
- Firebase Security Rules enforce the agreed permissions.
- Privileged database credentials are not exposed in the browser.

Tasks

- Configure authentication.
- Create the sign-in interface.
- Implement protected routes and session handling.
- Configure and test Firebase Security Rules.

US10 Use the Dashboard on Different Devices

User story

As a SAEON researcher working away from my desk, I want a dashboard that works on phones, tablets, and computers so that I can check station information wherever I am.

Acceptance criteria

- Key readings and controls remain readable on the tested screen sizes.
- Charts, filters, and navigation remain usable.
- Important information is not hidden by overlapping or clipped elements.

Tasks

- Implement responsive dashboard layouts.
- Adapt charts and tables for smaller screens.
- Test phone, tablet, and desktop layouts.
- Correct usability issues found during testing.

US11 Maintain Station Power

User story

As a station operator, I want a suitable battery and charging arrangement so that the station can continue monitoring between maintenance visits.

Acceptance criteria

- Battery, charging, and solar components are checked for compatibility.
- Operating and sleep-mode power consumption are measured.
- Charging behaviour is verified.
- Operating duration is tested against the team’s agreed requirement.

Tasks

- Assemble and test the power system.
- Integrate the solar charging arrangement.
- Implement and test low-power operation.
- Record power consumption and operating duration.

US12 Deploy and Validate the Station

User story

As a SAEON researcher, I want the assembled station tested in its deployment environment so that I can assess whether its measurements and communication are reliable.

Acceptance criteria

- Electronics are secured in the prepared enclosure.
- Sensor placement and station installation are documented.
- A 48-hour field test records communication, power behaviour, and measurement issues.
- Rainfall and temperature readings are compared with reference measurements.
- Results, faults, and deployment limitations are documented.

Tasks

- Complete final assembly and installation.
- Perform baseline calibration.
- Start the 48-hour field test.
- Collect manual reference measurements.
- Audit the results and record required improvements.

US13 Maintain the System

User story

As a future maintainer, I want clear technical documentation so that I can set up, troubleshoot, and maintain the station and dashboard.

Acceptance criteria

- Documentation covers components, wiring, configuration, and calibration.
- Firmware, cloud, database, and dashboard setup steps are included.
- Troubleshooting guidance and known limitations are recorded.
- Credentials are handled through documented configuration steps rather than published in the documentation.

Tasks

- Complete wiring and power documentation.
- Write firmware, database, and dashboard setup instructions.
- Document testing and troubleshooting.
- Review the README files for clarity and completeness.